AI Ray Tracer

Standard Operating Procedure: System Assessment

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Document Owner	Luke Neumann
System	AI Ray Tracer
Procedure Type	System Assessment
Supporting Workbook	AI Ray Tracer Assessment (.xlsx)

1. Document Control

1.1 Version Control

Date	Reason	Author	Version
7/27/26	Initial Write Up	Luke Neumann	1.0

1.2 Approval and Review

- The document owner shall review this SOP whenever the assessment method, rendering system, prompt set, or scoring workbook changes.
- A new version shall be issued when procedural changes could affect assessment results or comparisons between iterations.
- Minor spelling, formatting, or clarification changes may be documented as a minor revision.

2. Purpose

This SOP defines the standardized approach used to assess the AI Ray Tracer's image-generation performance and to capture structured evidence for improving the system. The procedure combines objective deliverable completion with human evaluation of visual appeal. Results are recorded by software iteration so engineers can compare performance, identify recurring failure modes, and provide structured assessment data to Codex or other development tools.

3. Scope

This SOP applies to all formal assessment runs of the AI Ray Tracer using the six benchmark prompts defined in this document. It covers test preparation, execution, render inspection, scoring, failure recording, spreadsheet entry, structured export, and final review. It does not replace unit testing, schema validation, performance profiling, or code review; those activities may be used as supporting evidence.

4. Roles and Responsibilities

Role	Responsibility
Operator	Runs each benchmark prompt under standard conditions, inspects the render, records actual deliverables, failed deliverable numbers, visual rating, and reasoning.
Assessment Owner	Maintains prompts, required deliverables, workbook formulas, and this SOP; reviews completed iterations.
Software Engineer	Uses structured assessment output to investigate failed deliverables, prioritize fixes, and verify changes in later iterations.

5. Required Resources

- AI Ray Tracer software build to be assessed.
- Access to the six benchmark prompts in Section 8.
- AI Ray Tracer Assessment workbook (.xlsx).
- Deliverables reference document for the six renders.
- A system capable of running the standard test configuration.
- A location for storing render outputs, logs, scene data, and exported assessment text.

6. Standard Test Conditions

Setting	Required Value
Quality	High (800 px)
Model Reasoning	Medium
Multi-Threading	Enabled

These settings shall remain unchanged for all six prompts within an iteration. If a setting must be changed, the deviation shall be recorded in the operator notes and the iteration shall not be compared directly with standard-condition iterations without qualification.

7. Process Flow

1. Select software version and create a new assessment iteration

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2. Confirm standard test conditions

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3. Run Prompt 1 through Prompt 6

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4. Save each render and related output

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5. Inspect each render against its required deliverables

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6. Record failed deliverable numbers

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7. Assign a visual appeal rating and reasoning

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8. Enter results into the workbook

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9. Review calculated scores

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10. Select the iteration on the Structured Export tab

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11. Copy the generated output for Codex or engineering analysis

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12. Review trends and define improvement actions

8. Benchmark Prompt Set

The six prompts are designed to assess immediate visual impact, material realism, feature clarity, structural complexity, grouped transformations, artistic range, and scene composition. Each prompt shall be submitted exactly as written unless the assessment specifically evaluates prompt revisions.

8.1 Glass, Chrome, and Reflections

Assessment focus: Showcases transparency, refraction, reflection, shadows, and varied materials.

Prompt: Create a dramatic studio still life on an infinite black-and-white checkered floor. Place a large transparent glass sphere in the center with high transparency and a realistic refractive index. Put a highly reflective dark chrome cube to its left and a polished gold closed cylinder to its right. Add two smaller colored spheres behind them. Use a low camera angle, strong perspective, one bright white point light above and to the left, deep shadows, crisp highlights, and a dark background. Keep the composition clean and cinematic.

8.2 Neon Planetary System

Assessment focus: Showcases emissive materials, bloom, spheres, scale, and depth.

Prompt: Create a cinematic miniature solar system floating over a dark reflective plane. Place one large glowing orange sun slightly off center with strong emissive color and bloom. Surround it with five planets of different sizes and colors, including a blue glass planet, a striped gas giant, and a small reflective red moon. Arrange the planets at different depths rather than in a straight line. Use a wide three-quarter camera angle, a subtle point light near the sun, a nearly black environment, vivid rim highlights, and strong but controlled bloom. Remember that emission is visually self-lit but does not illuminate nearby objects, so the requested point light helps sell the effect.

8.3 Pattern Material Gallery

Assessment focus: Showcases every supported procedural pattern in one readable image.

Prompt: Design a sophisticated material-gallery scene with five spheres displayed on individual matte pedestals above a neutral plane. Give the spheres these materials: black-and-white stripes, blue-to-gold gradient, red-and-cream rings, green-and-black checkers, and a perturbed purple-and-cyan pattern. Apply different scales and rotations to the patterns so their transformations are obvious. Use a centered gallery camera, one soft white point light above and to the left, balanced shadows, a charcoal background, and enough spacing that every sphere is clearly visible.

8.4 Geometric Sci-Fi Temple

Assessment focus: Showcases cubes, finite cylinders, triangles, grouping, and transformations.

Prompt: Create a symmetrical science-fiction temple built from grouped geometric shapes. Use two rows of tall closed cylinders as columns, stacked and rotated cubes for the central doorway, and large triangles forming a glowing emblem above it. Place everything on a slightly reflective dark plane. Use dark stone materials with subtle gold accents and a cyan emissive emblem with bloom. Frame the temple from a low, centered camera position looking slightly upward. Use one warm point light near the entrance and dramatic contrast while keeping the architecture readable.

8.5 Abstract Recursive Sculpture

Assessment focus: Showcases groups, nested transforms, repetition, and composition.

Prompt: Create an abstract floating sculpture made from three nested groups. Each group should contain a central reflective sphere, two rotated cubes, and three thin triangular fins. Scale and rotate each group differently so the full structure twists upward like a spiral. Use silver, deep blue, and copper materials. Suspend it above a matte plane with a soft circular-looking shadow. Use a three-quarter camera angle, a single bright point light from above, a dark navy background, and subtle blue emissive accents with restrained bloom.

8.6 Surreal Checkerboard Landscape

Assessment focus: Showcases planes, scale contrast, reflection, transparency, and visual storytelling.

Prompt: Create a surreal landscape on an infinite red-and-black checkered plane. Place three enormous glossy spheres receding toward the horizon, a transparent glass cube floating above the foreground, and several thin closed cylinders leaning at different angles like abstract towers. Add one small glowing cyan sphere as the focal point. Use an extremely low camera close to the ground, strong depth and scale, one pale point light high behind the camera, long shadows, reflective accents, and gentle bloom around the cyan sphere.

9. Required Deliverables by Render

The required deliverable count is standardized and shall not be changed by the operator during an assessment. A deliverable is counted as achieved only when it is clearly visible or otherwise verifiable in the completed render.

9.1 Glass, Chrome, and Reflections — 11 Required Deliverables

1. 1. Large transparent glass sphere centered in the scene.
2. 2. Visible refraction through the glass sphere.
3. 3. Dark reflective chrome cube on the left.
4. 4. Polished gold closed cylinder on the right.
5. 5. Two smaller colored spheres behind the main objects.
6. 6. Black-and-white checkered floor.
7. 7. Visible reflections on the chrome cube and gold cylinder.
8. 8. Deep shadows from the scene objects.
9. 9. Crisp highlights on the glass, chrome, and gold materials.
10. 10. Low camera angle with strong perspective.
11. 11. Dark, clean, cinematic background.

9.2 Neon Planetary System — 10 Required Deliverables

12. 1. Large glowing orange sun positioned slightly off center.
13. 2. Five planets with different sizes and colors.
14. 3. Blue transparent or refractive glass planet.
15. 4. Striped gas giant.
16. 5. Small reflective red moon.
17. 6. Planets arranged at different depths.
18. 7. Dark reflective plane below the planetary system.
19. 8. Visible point-light illumination near the sun.
20. 9. Strong but controlled bloom around the sun.
21. 10. Wide three-quarter camera angle with a nearly black background.

9.3 Pattern Material Gallery — 11 Required Deliverables

22. 1. Five clearly visible spheres.
23. 2. Five individual matte pedestals.
24. 3. Black-and-white striped sphere.
25. 4. Blue-to-gold gradient sphere.
26. 5. Red-and-cream ring-patterned sphere.
27. 6. Green-and-black checkered sphere.
28. 7. Perturbed purple-and-cyan patterned sphere.
29. 8. Visible differences in pattern scale.
30. 9. Visible differences in pattern rotation.
31. 10. Neutral plane with a charcoal background.
32. 11. Centered camera with balanced lighting and readable shadows.

9.4 Geometric Sci-Fi Temple — 10 Required Deliverables

33. 1. Symmetrical science-fiction temple structure.
34. 2. Two rows of tall closed-cylinder columns.
35. 3. Stacked and rotated cubes forming the central doorway.
36. 4. Large triangular emblem above the doorway.
37. 5. Dark stone materials across the main structure.
38. 6. Visible gold accents.
39. 7. Cyan emissive emblem with controlled bloom.
40. 8. Slightly reflective dark plane.
41. 9. Warm point light near the entrance.
42. 10. Low, centered camera angle looking slightly upward.

9.5 Abstract Recursive Sculpture — 12 Required Deliverables

43. 1. Three nested groups.
44. 2. Three reflective spheres, one per group.
45. 3. Six rotated cubes, two per group.
46. 4. Nine thin triangular fins, three per group.
47. 5. Different scale applied to each group.
48. 6. Different rotation applied to each group.
49. 7. Upward twisting or spiral composition.
50. 8. Silver material.
51. 9. Deep blue material.
52. 10. Copper material.
53. 11. Sculpture visibly floating above a matte plane with a soft shadow.
54. 12. Dark navy background with subtle blue emission and restrained bloom.

9.6 Surreal Checkerboard Landscape — 11 Required Deliverables

55. 1. Infinite red-and-black checkered plane.
56. 2. Three enormous glossy spheres.
57. 3. Spheres positioned to recede toward the horizon.
58. 4. Transparent glass cube floating above the foreground.
59. 5. Visible refraction through the glass cube.
60. 6. Multiple thin closed cylinders.
61. 7. Cylinders leaning at different angles.
62. 8. Small glowing cyan focal sphere.
63. 9. Extremely low camera angle close to the plane.

64. 10. Long shadows and reflective accents.
65. 11. Gentle bloom around the cyan sphere.

10. Full Testing Procedure

1. Create or select an iteration. Open the AI Ray Tracer Assessment workbook. Use the next available iteration tab or the iteration designated by the assessment owner.

2. Record build information. Enter the software version, assessment date, and operator name before executing prompts. The version must uniquely identify the tested build, release, or commit.

3. Confirm standard conditions. Set Quality to High (800 px), Model Reasoning to Medium, and Multi-Threading to Enabled. Confirm that no unapproved setting differs from the standard configuration.

4. Run the benchmark prompts. Submit each prompt in numerical order. Do not edit the wording between iterations unless the prompt itself is the subject of a controlled change.

5. Save assessment evidence. Save the final render for each prompt. When available, also retain the generated scene configuration, application log, and output filename associated with the render.

6. Perform initial visual review. View the full image at a practical inspection size. Confirm that the file opens correctly and that the render is complete enough to assess.

7. Assign visual appeal rating. Rate the render from 1 to 10 based on immediate visual impact, composition, readability, material appearance, artistic quality, and overall coherence. Enter a brief explanation supporting the rating.

8. Count actual deliverables. Review the numbered deliverables for that render. Count each deliverable only when it is clearly achieved. Enter the total in Actual Deliverables.

9. Record failed deliverables. Enter the numbers of deliverables that were not achieved as a comma-separated list, such as 1, 4, 3. Leave the field blank when all deliverables pass.

10. Review calculated scores. Confirm that the workbook calculates the Deliverable Score and Assessment Score. Do not manually overwrite calculated cells.

11. Repeat for all six renders. Complete the same inspection and entry process for every prompt in the iteration.

12. Review iteration summary. Verify that all six renders have complete inputs and review the overall iteration score and summary results.

13. Generate structured output. Open the Structured Export tab, select the completed iteration, and verify that the displayed version, scores, failed deliverables, and reasoning match the source iteration tab.

14. Copy for engineering analysis. Copy the generated structured output into Codex or the designated engineering analysis workflow. Use it to identify recurring failures, compare versions, and define corrective work.

15. Close the assessment. Store the workbook and evidence package in the designated project location. Record any deviations or unresolved concerns.

11. Assessment Method

11.1 Visual Appeal Rating

The operator assigns a whole-number rating from 1 to 10 and provides a concise explanation. The explanation should identify the strongest visual qualities and the most important weaknesses. The rating is a human-judgement measure and must not be inferred solely from deliverable completion.

Rating Range	General Interpretation
9-10	Exceptional visual quality; coherent, polished, and immediately compelling.
7-8	Strong result with minor visual weaknesses.
5-6	Acceptable but noticeably limited, inconsistent, or unpolished.
3-4	Weak result with major visual or compositional problems.
1-2	Unsuccessful or visually unusable result.

11.2 Objective Deliverable Score

The Deliverable Score measures the percentage of required deliverables achieved in the render:

Deliverable Score = Actual Deliverables ÷ Required Deliverables

Example: If a render achieves 8 of 10 required deliverables, the Deliverable Score is 0.80, or 80%.

11.3 Final Assessment Score

Each render receives one combined score using equal weighting between objective deliverable completion and visual appeal:

Assessment Score = 0.5 × (Deliverable Score × 100) + 0.5 × (Visual Appeal Rating × 10)

Example: A deliverable score of 80% and a visual appeal rating of 7 produce an assessment score of 75.00.

12. Data Entry Requirements

Field	Required Entry
Software Version	Unique version, build, release, or commit identifier.
Assessment Date	Date the iteration was executed.
Operator	Person performing the assessment.
Actual Deliverables	Whole number from zero through the required count for the render.
Failed Deliverable Numbers	Comma-separated deliverable identifiers, for example: 1, 4, 3.
Visual Appeal Rating	Whole number from 1 to 10.
Rating Reasoning	Brief explanation for the selected rating.
Calculated Fields	Deliverable Score and Assessment Score; these shall not be overwritten.

13. Structured Export Procedure

66. Complete all six render rows on the selected iteration tab.
67. Open the Structured Export tab.
68. Choose the completed iteration from the Select Iteration dropdown.
69. Verify the software version, assessment date, overall score, individual render scores, failed deliverables, and rating reasoning.
70. Copy the complete generated output block.

71. Paste the output into Codex or the approved engineering analysis prompt.
72. Preserve the output with the iteration evidence when it is used to make development decisions.

The structured output is intended to provide a consistent machine-readable summary of the assessment. It should be treated as an engineering input, not as a replacement for the source workbook or render evidence.

14. Acceptance and Completion Criteria

- All six prompts were executed under the documented test conditions.
- The software version, date, and operator are recorded.
- Each render contains an actual deliverable count, visual rating, and rating reasoning.
- Failed deliverable numbers are recorded when applicable.
- Calculated scores are present for all six renders.
- The selected iteration displays correctly on the Structured Export tab.
- Render evidence and the completed workbook are stored in the designated project location.

15. Deviations and Reassessment

Any deviation from the standard prompt wording, system settings, scoring method, required deliverables, or workbook formulas must be documented. A render may be rerun when the output is corrupted, incomplete because of a system error, or produced under incorrect conditions. The rerun reason should be recorded, and the original evidence should be retained when practical. Changes made to improve the software shall be assessed in a new iteration rather than overwriting a completed iteration.

16. Records and Retention

- Completed AI Ray Tracer Assessment workbook.
- Six final render images per iteration.
- Software version or commit identifier.
- Generated scene files and logs, when available.
- Structured export used for Codex or engineering analysis.
- Notes describing deviations, reruns, or significant findings.

17. External Resource Documents

Document Title	Owner	File Type
Deliverables	Luke Neumann	PDF
AI Ray Tracer Assessment	Luke Neumann	XLSX

Appendix A — Operator Quick Checklist

- ☐ Record software version, date, and operator.
- ☐ Confirm High (800 px), Medium reasoning, and multi-threading enabled.
- ☐ Run all six prompts exactly as written.
- ☐ Save each render.
- ☐ Rate visual appeal from 1 to 10 and explain the rating.
- ☐ Count achieved deliverables.
- ☐ Enter failed deliverable numbers as a comma-separated list.

- ☐ Verify calculated scores.
- ☐ Select the iteration on Structured Export.
- ☐ Copy the structured output for engineering analysis.
- ☐ Store the workbook and render evidence.